Course Materials

Chainsaw Maintenance & Cross Cutting

Course	Chainsaw Maintenance & Cross Cutting
Provider	Chainsaw Courses Ltd · chainsawcourses.com
Delivery	Fully Online — E-Learning (self-paced)
Hours	4 hrs GLH · 2 hrs Assessment · 4 hrs Self-Study · 10 hrs TQT
CPD	5 CPD Points (IIRSM)
Threshold	80% pass — randomised 45-question summative examination
Purpose	Training presentation & delegate materials

Contents

SECTION 1

How to Access the Live App & Admin Panel

All content unlocked for IIRSM assessors

The course is delivered through a custom-built e-learning platform. All video modules, quizzes, AI-assisted mock practice, summative examination, and certificate generation are accessible via the two routes below.

Admin / Assessor Access — All Content Unlocked

1. Go to app.chainsawcourses.com
2. Scroll to the bottom of the screen and click the 'Admin Panel' button.
3. Enter the admin credentials when prompted — Password: CHAIN26 — the full admin dashboard will appear.
4. Explore the dashboard tabs: Feedback, Backup, Storage, Policy Documents and more.
5. All Export buttons send data directly to protected Google Drive storage.
6. Click 'APP PREVIEW' in the top-right corner of the dashboard to enter the fully unlocked app.
7. All videos, features, certificates, and IIRSM-referenced content are fully accessible for review.

New Learner Journey — Standard Access Route

Experience the app exactly as a learner — sequential locking, quiz flow, and liability waiver. The app is under internal testing and has not been publicly released.

1. Log out of the admin panel (or open an incognito browser window).
2. Go to app.chainsawcourses.com and enter login code: CHAIN
3. Enter a name and email address of your choice.
4. You will be directed to the Liability Waiver — accept to enter the course.
5. Each module is locked until the previous video is fully watched and the quiz completed.
6. On completion, the 45-question exam unlocks. 80% pass auto-generates the IIRSM certificate.

SECTION 2

A Note on the Training Manual

Published author · File size & access options

Published Author

Author of 'The Chainsaw Manual' (Version 1.1, July 2026) — currently sold as a standalone physical learning aid to various colleges and training providers across the UK. The manual underpins the theoretical content of this eLearning course. The first edition was published in September 2024.

The training manual — Chainsaw Maintenance & Cross Cutting: A Comprehensive Technical Manual (v1.1) — is the primary delegate learning resource. It is a richly illustrated 138-page document covering all Learning Outcomes and Assessment Criteria, including Advanced Workshop Extension modules.

File Size Notice

The manual PDF is approximately 71 MB due to high-resolution diagrams and photographs. This exceeds most standard upload limits and cannot be attached directly to this IIRSM submission.

How to obtain the manual:

Option A — Request a Digital Copy via WeTransfer (Recommended)

Email info@chainsawcourses.com to request a digital copy. A download link will be sent via WeTransfer or a similar large-file transfer service.

Option B — View within the App

The full manual is embedded within the course platform. After logging in via the Admin Access route in Section 1, click APP PREVIEW. The manual is viewable within the app at any point during the course. All supporting policy documents are also accessible via the Admin Dashboard.

Option C — Request a Physical Copy

A physical copy of the manual can be requested by contacting David Daniel at chainsawcourses.com.

SECTION 3

Learning Outcome Framework

3 Units · 6 Learning Outcomes · 23 Assessment Criteria

The course is structured across three units and six formal learning outcomes, each broken into specific assessment criteria (ACs). All examination questions in Section 4 are tagged to their LO and AC. The modules listed under each outcome are the primary digital activities through which the content is delivered.

LO1 Legal Framework & Personal Safety

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Describe the statutory legal framework and personal safety requirements governing chainsaw operations.

AC 1.1 Identify employer and employee obligations under the Health and Safety at Work Act (HSWA) and global equivalents.
AC 1.2 Explain PUWER operational parameters governing tool maintenance and operator competence.
AC 1.3 Summarise COSHH control tracking required for hazardous fuels, battery cells, lubricants, and toxic flora species.
AC 1.4 Detail CE/UKCA and global standard class markings for safety helmets, hearing protection, gloves, and Type A/C protective trousers.

Activities: Equipment List · PPE & First Aid · Law & Legislation

LO2 Hazard Evaluation & Emergency Protocols

Unit 1 — Occupational Standards, Health & Safety, and Risk Evaluation

Evaluate environmental hazards, risk metrics, and implement emergency protocols for chainsaw site operations.

AC 2.1 Execute a 5-step site-specific risk assessment documenting ground hazards, pedestrian proximity, and structural vulnerabilities.
AC 2.2 Formulate an emergency communication and extraction map with grid references, postcodes, access limitations, and trauma kit deployments.
AC 2.3 Identify bio-security cleaning controls necessary to stop the spread of invasive arboreal pathogens and pests.

Activities: 5 Steps To Risk Assessment · Hazards & Risks · Emergency Planning Information

LO3 Chainsaw Architecture & Safety Features

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Analyse the mechanical differences, design attributes, and safety features of internal combustion and battery-powered chainsaws.

AC 3.1 Describe the 2-stroke combustion cycle and determine fuel-to-oil lubrication ratios at a standard 50:1 mix.
AC 3.2 Compare advantages and operational risks (including charging thermal runaway) of battery-powered vs. IC platforms.
AC 3.3 Map and explain the mechanical function of the 10 core safety features across the front, centre, and rear chainsaw architecture.

Activities: Chainsaw Safety Features · Battery Chainsaws

LO4 Diagnostic, Servicing & Maintenance Procedures

Unit 2 — Power Unit Architecture, Mechanical Integrity, and Component Maintenance

Describe the diagnostic, servicing, and maintenance procedures required to sustain the structural integrity of the chainsaw cutting assembly.

AC 4.1 Detail air filter cleaning procedures and interpret spark plug electrode colour indicators (Brown / Black / White-Grey).
AC 4.2 Explain safe carburettor adjusting rules using factory Idle (LA/T), Low (L), and High (H) screw limit constraints. (Advanced)
AC 4.3 Differentiate Rim and Spur drive sprockets and diagnose guidebar wear including burring, rail splaying, and thermal bluing.
AC 4.4 Identify chain pitch, gauge, and tooth shapes (Full-Chisel vs. Semi-Chisel) and calculate correct filing profile configurations.

Activities: Air Filter · Spark Plug · Cooling System · Exhaust · Fuel & Oil Filters · The Oiling System · Recoil Starter · Clutch Assembly · Sprocket · Chain Brake · Guidebar · Chain Basics · Chain Tension · How to identify a chainsaw chain · Replacing The Chain · Chain Sharpening · Kickback

LO5 Pre-Use Verification & Startup Methodologies

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Implement safe pre-use verification protocols and startup methodologies for safe chainsaw deployment.

AC 5.1 Differentiate between safe cold start floor-anchor and upright knee-clamp warm start methods for Husqvarna and Stihl platforms.
AC 5.2 Perform a 4-point dynamic check assessing chain brake engagement, oil dispersion flow, chain creep at tick-over, and off-switch motor cut.

Activities: Pre-Start Checks · Starting The Chainsaw · Pre-Use Checks

LO6 Cutting Mechanics: Tension & Compression

Unit 3 — System Startups, Operational Testing, and Processing Techniques

Apply mechanical principles to resolve tension and compression forces during timber cross-cutting operations.

AC 6.1 Analyse a log setup to determine where tension and compression forces reside.
AC 6.2 Explain the physics behind pulling chains versus pushing chains during cross-cutting.
AC 6.3 Define the bar tip kickback zone and explain how to execute precise plunge bore entries safely.
AC 6.4 Differentiate cut sequences for standard logs, oversized timber, and timber under extreme tension (Toast Rack / reduction sink).
AC 6.5 Evaluate specialised branch removal, snedding, and de-limbing sequences along a felled stem. (Advanced Extension)
AC 6.6 Identify site threats with windblown windfalls, establishing escape routes and managing root plate movements. (Advanced Extension)

Activities: Work Positioning · Cutting Basics · Tension & Compression · Releasing A Trapped Chainsaw · Bore Cutting · Oversized & Tensioned Timber · Stacking · Additional Cuts

SECTION 4

Syllabus Mapping Table

36 Modules · Learning Outcome & Assessment Criteria Alignment

The table below maps every active app module to its corresponding Learning Outcome (LO) and Assessment Criteria (AC). This mirrors the alignment table at the back of the training manual. Modules categorised as Course Requirements are reference documents and do not carry a formal LO/AC assignment.

#	Module Title	Type	Category	LO	Assessment Criteria
COURSE REQUIREMENTS — REFERENCE DOCUMENTS					
1	Equipment List	PDF	Course Req.	—	—
UNIT 1 — OCCUPATIONAL STANDARDS, HEALTH & SAFETY & RISK EVALUATION					
2	PPE & First Aid	Video	Assessment	LO1	AC 1.4
3	5 Steps To Risk Assessment	Video	Assessment	LO2	AC 2.1
4	Hazards & Risks	PDF	Assessment	LO1, LO2	AC 1.3, AC 2.3
5	Emergency Planning Information	Video	Assessment	LO2	AC 2.2
6	Law & Regulations	Video	Assessment	LO1	AC 1.1, AC 1.2
7	Chainsaw Safety Features	Video	Assessment	LO3	AC 3.3
UNIT 2 — POWER UNIT ARCHITECTURE, MECHANICAL INTEGRITY & COMPONENT MAINTENANCE					
8	Battery Chainsaws	Video	Maintenance	LO3	AC 3.2
9	Air Filter	Video	Maintenance	LO4	AC 4.1
10	Spark Plug	Video	Maintenance	LO4	AC 4.1
11	Cooling System	Video	Maintenance	LO4	AC 4.1
12	Exhaust	Video	Maintenance	LO4	AC 4.1
13	Fuel & Oil Filters	Video	Maintenance	LO4	AC 4.1, AC 3.1
14	The Oiling System	Video	Maintenance	LO4	AC 4.1
15	Recoil Starter	Video	Maintenance	LO4	AC 4.1
16	Clutch Assembly	Video	Maintenance	LO4	AC 4.1
17	Sprocket	Video	Maintenance	LO4	AC 4.3
18	Chain Brake	Video	Maintenance	LO3	AC 3.3
19	Guidebar	Video	Maintenance	LO4	AC 4.3
20	Chain Basics	Video	Maintenance	LO4	AC 4.4
21	Chain Tension	Video	Maintenance	LO4	AC 4.4
22	How to identify a chainsaw chain	Video	Maintenance	LO4	AC 4.4
23	Replacing The Chain	Video	Maintenance	LO4	AC 4.4
24	Chain Sharpening	Video	Maintenance	LO4	AC 4.4
UNIT 3 — SYSTEM STARTUPS, OPERATIONAL TESTING & PROCESSING TECHNIQUES					
25	Kickback	Video	Cross Cutting	LO6	AC 6.3
26	Work Positioning	Video	Cross Cutting	LO6	AC 6.1
27	Pre-Start Checks	Video	Cross Cutting	LO5	AC 5.2
28	Starting The Chainsaw	Video	Cross Cutting	LO5	AC 5.1
29	Pre-Use Checks	Video	Cross Cutting	LO5	AC 5.2
30	Cutting Basics	Video	Cross Cutting	LO6	AC 6.1
31	Tension & Compression	Video	Cross Cutting	LO6	AC 6.1, AC 6.2
32	Releasing A Trapped Chainsaw	Video	Cross Cutting	LO6	AC 6.2
33	Bore Cutting	Video	Cross Cutting	LO6	AC 6.3
34	Oversized & Tensioned Timber	Video	Cross Cutting	LO6	AC 6.4

#	Module Title	Type	Category	LO	Assessment Criteria
35	Stacking	Video	Cross Cutting	LO6	AC 6.4
36	Additional Cuts	Video	Cross Cutting	LO6	AC 6.4

Key

LO1–LO6	Learning Outcomes 1–6 across three units of the qualification
AC x.x	Assessment Criteria — each LO has 2–6 specific criteria
—	Course Requirements modules are prerequisite reference docs; no formal LO/AC assigned
Video	Interactive video module with embedded quiz (sequentially gated)
PDF	Read-only reference document (does not gate progression)

SECTION 5

Compliance Tools & Practical Worksheets

Available within the course app · PPE Verification · Risk Assessment · Emergency Action Plan · Bio-Security · CPD News

All practical compliance worksheets are embedded directly within the course app and are completed digitally by the learner. This approach ensures records are date-stamped, stored securely, and immediately exportable as a PDF or plain-text report for portfolio submission or regulatory inspection.

PPE Verification Checklist

Accessible from the app Inspection Checklist. The learner confirms each item of PPE — helmet, hearing protection, gloves, trousers, boots, hi-vis, and first aid kit — against its UK/international standard (EN 397, EN 352, EN ISO 11393, EN ISO 17249, EN ISO 20471, BS 8599-1). Mapped to LO1 / AC 1.4 and LO2 / AC 2.2.

Pre-Start & Pre-Use Chainsaw Inspection

Also within the app Inspection Checklist. Covers chain tension, sharpness, brake function, bar condition, oiler, fuel mix, air filter, handguards, chain catcher, exhaust, anti-vibration mounts, and controls. A 4-point dynamic check confirms brake engagement, oil spray, chain creep at idle, and off-switch cut. Mapped to LO5 / AC 5.2.

Risk Assessment, Emergency Action Plan & Bio-Security Log

A 5-step site risk assessment template, Emergency Action Plan (site grid reference, nearest A&E, first aider, muster point), and bio-security cleaning log are available via the app dashboard. Each record is retained and can be exported for IIRSM portfolio use. Mapped to LO2 / AC 2.1, 2.2, and 2.3.

News & CPD Updates

Industry news is published regularly through the app and tagged to its relevant Learning Outcome and AC. Engaging with updates directly supports the IIRSM RM&CF Ongoing Competence Assurance pillar. Accessible via the Admin Dashboard under the Updates tab.

SECTION 6

Assessment Bank

151 Multiple-Choice Questions · Summative Examination Pool

How the exam works

The summative examination draws a randomised 45 questions from this 151-question bank. Questions are mapped to their Learning Outcome (LO) and Assessment Criterion (AC). Learners must achieve 80% or above to pass. Correct answers are highlighted in orange with ' .

Q1 LO1 · AC 1.4

What is the PRIMARY purpose of chainsaw protective trousers?

A. Provide warmth in cold conditions
B. Stop or slow a chainsaw chain on contact with the leg '
C. Improve operator visibility
D. Protect against thorns and branches only

Q2 LO1 · AC 1.4

What is the FIRST action to take when a chainsaw operator sustains a deep cut?

A. Apply direct pressure to control bleeding and call 999 '
B. Remove the chainsaw chain from the wound immediately
C. Give the casualty food and water
D. Continue working and treat the injury later

Q3 LO1 · AC 1.4

Which standard governs chainsaw protective clothing in the UK?

A. BS EN 381 '
B. BS EN 397
C. BS EN 471
D. BS EN 166

Q4 LO1 · AC 1.4

How frequently must chainsaw PPE be inspected for damage?

A. Once per year
B. Once per month
C. Before every use '
D. Only when it visibly looks worn

Q5 LO2 · AC 2.1

What is the FIRST step in the 5-step risk assessment process?

A. Record your findings
B. Identify the hazards '
C. Evaluate the risks
D. Implement control measures

Q6 LO2 · AC 2.1

Under the Management of Health and Safety at Work Regulations 1999, who must carry out a risk assessment?

A. The Health and Safety Executive
B. The local authority
C. Employers and self-employed persons '
D. Only qualified safety officers

Q7 LO2 · AC 2.1

A risk assessment must be reviewed when:

A. A new calendar year begins
B. Working conditions change significantly '
C. A manager verbally requests it
D. Staff turnover occurs

Q8 LO2 · AC 2.1

In a risk assessment, what does 'risk' specifically mean?

A. The likelihood that harm will occur from a hazard '
B. The physical object that could cause harm
C. The financial cost of an accident
D. The legal penalties for non-compliance

Q9 LO2 · AC 2.2

Before starting chainsaw work, what emergency information must be established?

A. The nearest fuel station
B. The nearest hardware store
C. The site manager's home address
D. The location of emergency services and a means to contact them '

Q10 LO2 · AC 2.2

What is the recommended approach for lone workers operating a chainsaw?

A. Check in every 4 hours
B. Check in every 8 hours
C. Check in at regular agreed intervals (e.g. every 30 minutes) '
D. Check in once per day

Q11 LO2 · AC 2.2

An emergency action card should include:

A. Personal social media contacts
B. Equipment hire costs
C. Site location, grid reference, and emergency service numbers '
D. Manufacturer warranty information

Q12 LO2 · AC 2.2

What does 'point of work risk assessment' mean?

A. An assessment carried out at the actual working location before starting work '
B. An assessment completed in the office before site arrival
C. A financial assessment of the job
D. An assessment completed after work is finished

Q13 LO1 · AC 1.1, AC 1.2

Which legislation requires work equipment to be safe and suitable for its intended use?

A. Health and Safety at Work Act 1974
B. Workplace (Health, Safety and Welfare) Regulations 1992
C. Provision and Use of Work Equipment Regulations 1998 (PUWER) '
D. Manual Handling Operations Regulations 1992

Q14 LO1 · AC 1.1, AC 1.2

Who is the primary enforcing authority for health and safety in UK forestry and arboriculture?

A. The local council
B. The Forestry Commission
C. The Environment Agency
D. The Health and Safety Executive (HSE) '

Q15 LO1 · AC 1.1, AC 1.2

The Health and Safety at Work Act 1974 places a duty on employees to:

A. Take reasonable care for their own and others' health and safety '
B. Purchase their own PPE
C. Report all incidents to the police
D. Only follow instructions given in writing

Q16 LO1 · AC 1.1, AC 1.2

PUWER stands for:

A. Personal Use of Work Equipment Regulations
B. Protection and Upkeep of Work Equipment Regulations
C. Provision and Use of Work Equipment Regulations '
D. Prevention of Unsafe Work Equipment Risks

Q17 LO3 · AC 3.3

What is the primary function of the chain brake?

A. Stop the chain rotating rapidly in the event of kickback '
B. Sharpen the chain automatically
C. Prevent the engine from starting accidentally
D. Control engine speed at idle

Q18 LO3 · AC 3.3

The chain catcher on a chainsaw is designed to:

A. Collect sawdust
B. Hold the chain during sharpening
C. Reduce chain noise
D. Protect the operator if the chain breaks or derails '

Q19 LO3 · AC 3.3

What is the purpose of the anti-vibration (AV) system on a chainsaw?

A. Reduce engine noise emissions
B. Prevent chain derailment
C. Reduce vibration transmitted from the saw to the operator's hands '
D. Improve fuel efficiency

Q20 LO3 · AC 3.3

The front hand guard on a chainsaw:

A. Protects the operator's hand from the chain and activates the chain brake '
B. Acts as a handhold during transport
C. Supports the weight of the saw
D. Prevents fuel from splashing the operator

Q21 LO4 · AC 4.1

Why is it critical to clean or replace a blocked chainsaw air filter?

A. A blocked filter makes the saw louder
B. A blocked filter improves chain speed
C. A blocked filter has no effect on performance
D. A blocked filter causes a rich mixture, poor performance and overheating '

Q22 LO4 · AC 4.1

How should a foam air filter be cleaned?

A. Washed in warm soapy water, dried fully before re-fitting '
B. Washed in petrol and left to dry
C. Scraped clean with a metal tool
D. Replaced immediately without attempting to clean

Q23 LO4 · AC 4.1

When inspecting the air filter, you should also check:

A. The bar sprocket nose
B. The spark plug gap only
C. The filter housing and gasket for damage or leaks '
D. The chain tension only

Q24 LO4 · AC 4.1

Operating with a damaged or missing air filter will most likely cause:

A. Improved engine performance
B. Lower chain speed only
C. Reduced vibration only
D. Increased engine wear due to ingested debris '

Q25 LO4 · AC 4.1

What should you assess when inspecting a chainsaw spark plug?

A. Electrode condition, gap, and look for fouling or damage '
B. Replace it after every use regardless of condition
C. Only inspect if the saw fails to start
D. Clean it with water and refit

Q26 LO4 · AC 4.1

A black, sooty spark plug electrode typically indicates:

A. A lean fuel mixture (too much air)
B. A rich fuel mixture or excess oil in the combustion chamber '
C. Correct engine operation
D. Severe overheating

Q27 LO4 · AC 4.1

What tool is needed to correctly set the spark plug electrode gap?

A. A torque wrench
B. A vernier calliper
C. A multimeter
D. A feeler gauge '

Q28 LO4 · AC 4.1

What is the risk of overtightening a spark plug during installation?

A. The plug may fall out during operation
B. The engine will run too rich
C. The cylinder head threads can be stripped or the plug body cracked '
D. The chain will derail

Q29 LO4 · AC 4.1

Why must cooling fins on a chainsaw cylinder be kept clean?

A. To improve the appearance of the saw
B. To reduce chain noise
C. To improve fuel efficiency only
D. To allow adequate airflow and prevent engine overheating '

Q30 LO4 · AC 4.1

What should be used to clean clogged cooling fins?

A. A soft brush or plastic tool to dislodge debris without damaging fins '
B. A metal scraper
C. High-pressure water directly on the engine
D. Petrol sprayed directly into the fins

Q31 LO4 · AC 4.1

Where does cooling air typically enter most chainsaw designs?

A. Through the exhaust port
B. Through the chain tensioner slot
C. Through the flywheel and air intake shroud '
D. Through the bar mounting holes

Q32

LO4 · AC 4.1

Operating with blocked cooling fins can result in:

A. Better fuel economy
B. Increased chain speed
C. Reduced vibration levels
D. Piston seizure and major engine damage '

Q33

LO4 · AC 4.1

What hazard does a blocked spark arrestor in the exhaust present?

A. Increased fuel consumption only
B. Reduced chain speed
C. Fire risk from ejected hot carbon particles '
D. Loss of chain lubrication

Q34

LO4 · AC 4.1

Signs of a failing chainsaw exhaust system include:

A. Quieter engine operation
B. Improved fuel economy
C. Reduced chain vibration
D. Exhaust leaks causing burn risk and increased operator noise exposure '

Q35

LO4 · AC 4.1

When should the spark arrestor be cleaned or replaced?

A. Per the manufacturer's schedule or when visibly blocked '
B. It is a sealed unit and never serviced
C. Only when the saw fails to start
D. After every single use

Q36

LO4 · AC 4.1

Chainsaw exhaust fumes are particularly dangerous in enclosed or confined spaces because:

A. They are brightly coloured and have a strong smell
B. They cause skin irritation only
C. Carbon monoxide is colourless and odourless and can reach lethal concentrations '
D. They corrode metal parts rapidly

Q37

LO4 · AC 4.1, AC 3.1

What is the purpose of the chainsaw fuel filter?

A. To mix fuel and bar oil
B. To store reserve fuel
C. To heat the fuel before combustion
D. To prevent debris entering the fuel system and damaging the carburettor '

Q38

LO4 · AC 4.1, AC 3.1

Bar and chain oil should have what key property to be effective?

A. Sufficient tackiness to adhere to the chain and bar during operation '
B. Be very thin to flow freely at all temperatures
C. Be identical in specification to engine oil
D. Be water-based for environmental compliance

Q39

LO4 · AC 4.1, AC 3.1

A partially blocked fuel filter will typically cause:

A. Improved fuel economy
B. Overfuelling and a rich mixture
C. Fuel starvation, causing the engine to cut out under load '
D. No noticeable change in performance

Q40 LO4 · AC 4.1, AC 3.1

How often should chainsaw fuel and bar oil filters be inspected?

A. Only during the annual service
B. Only when a fuel problem is noticed
C. Every 1,000 hours of use
D. Per the manufacturer's maintenance schedule '

Q41 LO4 · AC 4.1

What must be confirmed before starting a chainsaw using the recoil starter?

A. Chain brake is engaged, saw is secure, and area is clear of people '
B. Chain is loose for easy starting
C. Throttle is fully open
D. Bar oil reservoir is empty

Q42 LO4 · AC 4.1

If the recoil starter cord does not retract after pulling, this most likely indicates:

A. The engine is flooded
B. A broken or damaged recoil spring '
C. The chain brake is disengaged
D. Low fuel level

Q43 LO4 · AC 4.1

What is the correct ground-starting position for a chainsaw?

A. Hold the saw between your knees with the bar pointing forward
B. Balance the saw on the chain brake guard
C. Place the rear handle under your right foot, hold the front handle, bar clear of the ground '
D. Hold the saw by the bar with one hand

Q44 LO4 · AC 4.1

The chain brake must be engaged during the starting procedure:

A. Only after the engine has warmed up
B. Never — it prevents the engine from starting
C. Only when starting on steep slopes
D. At all times throughout the starting procedure '

Q45 LO4 · AC 4.1

When does the centrifugal clutch on a chainsaw typically engage the chain?

A. At idle speed only
B. Immediately on engine start
C. At a higher RPM as engine speed increases above idle '
D. Only when the throttle trigger is fully released

Q46 LO4 · AC 4.1

A worn clutch drum will most commonly result in:

A. The chain running at idle — a serious safety hazard '
B. Improved chain engagement at lower speeds
C. Reduced fuel consumption
D. Increased cutting speed

Q47 LO4 · AC 4.1

What is typically required to safely remove a chainsaw clutch assembly?

A. A standard socket set only
B. Mole grips on the flywheel
C. Only a large screwdriver
D. A clutch removal tool with the correct thread direction and a piston stop '

Q48 LO4 · AC 4.1

The clutch springs are responsible for:

- **A. Holding the clutch pads retracted at idle so the chain does not move '**
- B. Returning the throttle trigger to idle position
- C. Tensioning the guide bar
- D. Driving the oil pump

Q49 LO4 · AC 4.3

Why must a worn drive sprocket be replaced?

- A. To improve the appearance of the saw
- B. A worn sprocket reduces vibration
- C. Worn teeth improve chain engagement
- **D. To decrease wear on the chain and guidebar '**

Q50 LO4 · AC 4.3

When fitting a new chain, what else should always be inspected for wear?

- **A. The drive sprocket, as chain and sprocket wear together '**
- B. The air filter
- C. The spark plug
- D. The fuel filter

Q51 LO4 · AC 4.3

What is the maximum recommended sprocket wear before it must be replaced?

- A. 0.1mm
- B. 1.0mm
- **C. 0.5mm or as specified by the manufacturer '**
- D. 2.0mm

Q52 LO4 · AC 4.3

A rim sprocket design allows:

- A. Greater cutting accuracy
- B. Higher idle speeds
- C. Direct drive to the chain without a clutch
- **D. The rim sprocket can be replaced without replacing the clutch drum as well '**

Q53 LO3 · AC 3.3

The inertia-activated chain brake operates by:

- A. The operator pressing a dedicated button
- B. The front hand guard being pressed by the operator's wrist
- **C. A flyweight mechanism detecting rapid saw body rotation and triggering automatically '**
- D. Releasing the throttle trigger

Q54 LO3 · AC 3.3

How should the chain brake be tested in the workshop environment?

- A. Rev to full throttle and deliberately trigger kickback
- B. Check it in the workshop with the bar removed
- C. It does not require testing before use
- **D. Reassemble the saw and activate the chain brake and try to spin the chain manually '**

Q55 LO3 · AC 3.3

If the chain continues rotating when the chain brake is activated, you must:

- **A. Stop work immediately — do not use the saw until the brake is repaired '**
- B. Continue working but exercise extra care
- C. Increase the engine idle speed
- D. Tighten the guide bar nuts

Q56 LO3 · AC 3.3

What should be checked when maintaining the chain brake?

A. Oil the brake band lightly after each use to prevent corrosion
B. The brake band must be clean and free of oil or grease — replace if worn, glazed, or below minimum thickness '
C. The brake spring should be removed during long-term storage
D. Tighten the front hand guard pivot to increase braking force

Q57 LO4 · AC 4.3

Why should the guide bar be regularly rotated 180 degrees?

A. To improve chain speed
B. To change the cutting direction
C. To allow even wear on both sides of the bar rail '
D. To clear the bar oil feed holes

Q58 LO4 · AC 4.3

What should be checked in the guide bar groove during inspection?

A. Fuel tank level
B. Bar length only
C. Drive link count
D. Groove depth, rail burrs, and bar oil holes for blockage '

Q59 LO4 · AC 4.3

A guide bar with a pinched (closed or damaged) rail groove causes:

A. Reduced chain lubrication and premature bar and chain wear '
B. Improved cutting accuracy
C. Faster chain speed
D. Reduced vibration

Q60 LO4 · AC 4.3

Bar oil feed holes must be kept clear to ensure:

A. Correct chain tension is maintained
B. Correct chain brake function
C. Adequate lubrication of the chain and bar throughout cutting '
D. Engine cooling airflow

Q61 LO4 · AC 4.4

Drive links on a chainsaw chain serve to:

A. Cut through the wood
B. Reduce vibration transmission
C. Measure the chain length
D. Engage with the drive sprocket and guide bar groove to drive and guide the chain '

Q62 LO4 · AC 4.4

What is a semi-chisel chain profile good for?

A. Cutting timber under tension
B. Cutting mostly softwood with increased kickback
C. Cutting mostly hardwood with reduced kickback '
D. Cutting metal and other foreign objects

Q63 LO4 · AC 4.4

The depth gauge (raker) on a chain cutter controls:

A. Chain speed
B. Chain tension
C. The depth of cut each cutter tooth takes into the wood '
D. Bar lubrication flow

Q64 LO4 · AC 4.4

Which part of the chainsaw chain actually cuts the wood?

A. The tie straps
B. The drive links
C. The left and right cutter teeth '
D. The bumper drive links

Q65 LO4 · AC 4.4

How is correct chain tension identified on a standard chainsaw?

A. The chain hangs loosely below the bar
B. The chain cannot be moved by hand at all
C. The chain pulls slightly away from the bar but snaps back with drive links remaining in the groove '
D. Pliers are needed to move the chain along the bar

Q66 LO4 · AC 4.4

Why is operating with a chain that is too loose dangerous?

A. It reduces fuel consumption
B. It improves cutting performance
C. It reduces operating noise
D. A loose chain can derail from the bar, creating a serious injury risk '

Q67 LO4 · AC 4.4

When should chain tension be checked?

A. Before use and at regular intervals — new chains need especially frequent re-tensioning '
B. Only at the start of the working day
C. Only when the chain appears loose
D. Once per week only

Q68 LO4 · AC 4.4

A chain that is too tight will cause:

A. Improved cutting speed
B. Excessive wear on bar, chain, and sprocket, and potentially dangerous bar heating '
C. Reduced kickback risk
D. Better lubrication distribution

Q69 LO4 · AC 4.4

How do you identify a Stihl chainsaw chain?

A. Check the colour of the tie straps and cutter teeth
B. Count the number of cutters and divide by two
C. Find the number below the depth gauge and on the drive link, then cross-reference with the chain chart '
D. Look for the Stihl logo stamped on the guide bar

Q70 LO4 · AC 4.4

Where on a chainsaw chain is the identification number stamped?

A. On the tie straps
B. On the depth gauge only
C. On the drive link or on the cutting tooth below the depth gauge '
D. On the cutter tooth face only

Q71 LO4 · AC 4.4

What does the chain identification number allow you to do?

A. Set the correct chain tension
B. Determine the bar oil viscosity
C. Identify the pitch, gauge and which file size and filing angle to use '
D. Calculate the correct bar oil viscosity

Q72 LO4 · AC 4.4

How do you identify a Husqvarna chainsaw chain?

A. By the number of cutting teeth on the top of the chain
B. By the colour of the depth gauge
C. By the chain weight in grams
D. Find the part number stamped on the drive link and cross-reference with the chain identification chart '

Q73 LO4 · AC 4.4

What is the pitch of a chainsaw chain?

A. The overall length of the chain in inches
B. The width of the drive link that fits the guide bar groove
C. The average distance between three consecutive rivets divided by two '
D. The angle of the cutting tooth top plate

Q74 LO4 · AC 4.4

What is the gauge of a chainsaw chain?

A. The number of drive links in the chain
B. The thickness of the drive link that fits into the guide bar groove '
C. The distance between consecutive rivets
D. The height of the depth gauge above the cutter

Q75 LO4 · AC 4.4

What are the two main cutter profiles used on chainsaw chains?

A. Round and square
B. Semi-chisel and full-chisel '
C. Micro and macro chisel
D. Left and right profile only

Q76 LO4 · AC 4.4

When selecting a replacement chain, which three measurements must match the original?

A. Bar length, engine displacement, and chain weight
B. Chain colour, manufacturer, and number of cutters
C. Pitch, gauge, and number of drive links '
D. Cutter height, depth gauge setting, and chain age

Q77 LO4 · AC 4.4

When sharpening a cutter, the file should be held at:

A. The manufacturer-specified horizontal angle (typically 25-35 degrees) '
B. 90 degrees straight across the chain
C. 45 degrees downward into the cutter
D. Parallel to the guide bar

Q78 LO4 · AC 4.4

The depth gauge (raker) on a chain cutter:

A. Guides the file during sharpening
B. Measures chain tension directly
C. Indicates when the chain needs full replacement
D. Limits how deeply each cutter bites into the wood, controlling chip thickness '

Q79 LO4 · AC 4.4

If depth gauges become too high relative to the cutters after repeated sharpening:

A. The chain cuts more aggressively
B. Chain speed increases significantly
C. The chain cuts poorly because the cutters cannot engage the wood adequately '
D. There is no effect on cutting performance

Q80 LO4 · AC 4.4

After sharpening, all cutters on the chain must be:

A. Different lengths to compensate for uneven wood hardness
B. The same length and sharpened to the same angle for smooth, efficient cutting '
C. Longer on one side than the other
D. Set to varying depth gauge heights

Q81 LO6 · AC 6.4

What is the maximum recommended height for a freestanding timber stack without additional support?

A. 1.0 metre, as specified in the site risk assessment '
B. 3.0 metres
C. There is no recommended maximum height
D. 1.5 metres

Q82 LO6 · AC 6.4

When stacking multiple different species, what should you do?

A. Stack them randomly to save time
B. Arrange and separate species into their respective stacks '
C. Mix species to improve stability
D. Only stack one species per day

Q83 LO6 · AC 6.4

The stacking work area should be:

A. Uneven ground to give better grip to stacked timber
B. As close to felling operations as possible regardless of slope
C. On stable, clear ground away from slope run-out zones '
D. Directly beside any active felling work

Q84 LO6 · AC 6.4

Before approaching an existing timber stack for further work, you must:

A. Approach quickly to check for hazards
B. Place additional timber on top to test stability
C. Ignore the stack and focus on other nearby work
D. Visually assess the stack for signs of instability before approaching '

Q85 LO6 · AC 6.1

The recommended cutting stance for cross-cutting timber is:

A. Feet together, facing the cut directly
B. Kneeling beside the log
C. Balanced stance with feet shoulder-width apart, good footing, body weight evenly distributed '
D. Standing directly over the log

Q86 LO6 · AC 6.1

Why must the operator never stand directly in line with the bar when cross-cutting?

A. It reduces cutting efficiency
B. It is harder to see the cut line accurately
C. It increases vibration exposure to the legs
D. In the event of kickback, the bar swings towards the operator's face and upper body '

Q87 LO6 · AC 6.1

When should the operator plan and check their escape route?

A. Before starting any cut '
B. After the cut is completed
C. Only when working on steep slopes
D. Only when working in proximity to other people

Q88

LO6 · AC 6.1

Why do you need to keep your thumb wrapped around the handle?

A. To reduce hand fatigue during long cuts
B. To provide a firm secure grip '
C. To activate the throttle lock more easily
D. To improve vibration dampening

Q89

LO5 · AC 5.2

Which of the following is NOT typically part of a pre-start chainsaw check?

A. Checking fuel and bar oil levels
B. Inspecting chain brake function
C. Adjusting the carburettor idle speed '
D. Checking chain tension and sharpness

Q90

LO5 · AC 5.2

If fuel is leaking from the saw during a pre-start check, you should:

A. Stop — do not start the saw and arrange repair before use '
B. Continue but check it again at lunch
C. Use the saw quickly before the leak worsens
D. Wipe the fuel away and proceed normally

Q91

LO5 · AC 5.2

Pre-start checks should be carried out:

A. Once per week
B. Only when the operator changes
C. Before each work session and after the saw has been left unattended '
D. Only at the start of a new contract

Q92

LO5 · AC 5.2

During a pre-start chain inspection, you are checking for:

A. Number of cutting teeth only
B. Chain colour and manufacturer markings
C. Chain speed at idle
D. Sharpness, correct tension, drive links in groove, and no damaged links '

Q93

LO5 · AC 5.1

Why must the chain brake be engaged before starting a chainsaw?

A. It is not necessary to engage the brake for starting
B. To increase compression for easier starting
C. To warm the engine before operational use
D. To prevent chain rotation during the starting procedure, reducing injury risk '

Q94

LO5 · AC 5.1

The correct ground-starting position for a chainsaw is:

A. Place rear handle under right foot, hold front handle with bar clear of ground '
B. Hold front handle, place saw between the knees
C. Hang saw by the front guard and pull cord upward
D. Start with the bar held vertically

Q95

LO5 · AC 5.1

A flooded chainsaw engine is typically caused by:

A. Too little fuel reaching the engine
B. A blocked air filter
C. Excess fuel in the engine from incorrect choke or throttle use during starting '
D. Low engine compression

Q96 LO5 · AC 5.1

After starting a cold engine, why must the choke be returned to the run position?

A. The choke improves fuel efficiency during warm-up
B. Leaving the choke on causes an over-rich mixture that stalls or damages the engine '
C. The choke controls chain speed at idle
D. The choke only affects the initial starting phase

Q97 LO5 · AC 5.2

The throttle trigger interlock must be confirmed to:

A. Open the throttle smoothly when fully pressed
B. Rev the engine to maximum speed
C. Prevent throttle engagement unless the interlock is simultaneously depressed '
D. Allow the chain to move at idle speed

Q98 LO5 · AC 5.2

How is bar oil reaching the chain confirmed during a pre-use check?

A. Confirm the bar oil reservoir is completely empty before refilling
B. Remove the oil cap during operation to observe flow
C. Run the chain dry for the first 10 minutes and then check
D. Hold saw over light-coloured material and check for oil spray while revving '

Q99 LO5 · AC 5.2

Why should you check the chain is static whilst idling?

A. To confirm the chain brake is not engaged
B. To ensure the chain is not moving unintentionally '
C. To measure idle speed accuracy
D. To check bar oil is reaching the chain

Q100 LO5 · AC 5.2

If the chain brake does NOT stop the chain when tested during a pre-use check, you must:

A. Continue working but remain extra vigilant
B. Tighten the guide bar nuts and retest
C. Take the saw out of service immediately and arrange for repair '
D. Engage the brake manually with your hand as a workaround

Q101 LO6 · AC 6.1

What is the difference between a pushing chain and a pulling chain?

A. A pushing chain cuts on the upstroke; a pulling chain cuts on the downstroke
B. The pulling chain (bottom of bar) draws the saw into the cut; the pushing chain (top of bar) pushes the saw back towards the operator '
C. Pushing chains are used on petrol saws; pulling chains on battery saws
D. There is no difference — all chainsaw chains work identically

Q102 LO6 · AC 6.1

Why is it important to secure timber before cross-cutting?

A. To improve the visual accuracy of the cut line
B. To prevent the saw from overheating during the cut
C. To reduce the need for bumper spike contact
D. To ensure the timber cannot roll or move unexpectedly, which could trap the bar or endanger the operator '

Q103 LO6 · AC 6.1

What happens if the chain contacts the ground or a foreign object during cutting?

A. The saw engine stalls automatically as a safety measure
B. The chain brake engages to protect the operator
C. The chain will immediately become blunt or damaged, reducing cutting efficiency and increasing hazard '
D. Chain speed increases due to reduced resistance

Q104 LO6 · AC 6.1

When making a standard downward cross-cut, which part of the bar should you use?

A. The nose of the bar for maximum reach
B. The upper edge of the bar (pushing chain)
C. Either edge — it makes no difference
D. The lower edge of the bar (pulling chain) to draw the saw through the timber in a controlled manner '

Q105 LO6 · AC 6.1, AC 6.2

When a log is in compression on its upper face (supported at both ends), the safe cutting technique is:

A. Cut straight through from the top in a single pass
B. Cut the compression side first (from the top), then finish the cut from below '
C. Cut from the end of the log inward
D. Do not attempt to cut logs in compression

Q106 LO6 · AC 6.1, AC 6.2

When a log is in tension on its upper face (cantilevered, supported at one end), the safe technique is:

A. Cut the top (tension side) first, then finish from below
B. Cut straight through from the top in a single pass
C. Cut the compression side first (from below), then finish the cut from the top '
D. Always cut from the side to avoid all risk

Q107 LO6 · AC 6.1, AC 6.2

If a bar becomes pinched in a compression cut, what is the likely outcome?

A. The chain cuts faster to free itself
B. The chain brake activates automatically
C. The saw will safely shut down
D. The wood grips the bar, potentially stalling the saw or causing loss of control '

Q108 LO6 · AC 6.1, AC 6.2

Before any cross-cut, the operator must assess:

A. Where tension and compression zones are to determine the safe cutting sequence '
B. The colour and species of the timber only
C. Only whether the timber is wet or dry
D. Only the diameter of the timber

Q109 LO6 · AC 6.2

If the chainsaw bar becomes pinched during a cut, the FIRST action is:

A. Pull the saw repeatedly until it comes free
B. Rev the engine to maximum to force the bar free
C. Remove the chain from the bar while it remains pinched
D. Turn the saw off and assess the situation '

Q110 LO6 · AC 6.2

A safe method to release a pinched bar without power tools is:

A. Lever the kerf open using a wedge or felling lever inserted into the cut '
B. Strike the log firmly with the chainsaw body
C. Cut from a completely different direction at full speed without further assessment
D. Stand on the log and pull the saw upward with both hands

Q111 LO6 · AC 6.2

When using a felling wedge to release a trapped bar, it should be inserted:

A. Into the kerf on the compression side to open the cut '
B. Into the kerf on the tension side
C. Using the chainsaw bar itself to drive the wedge in
D. While the chain is still running at full speed

Q112 LO6 · AC 6.2

If the engine stalls while the bar is pinched in timber, you should:

A. Restart the engine immediately and force the cut through
B. Use bare hands to manually open the kerf
C. Make the saw safe and use hand tools or a second saw to release it safely '
D. Leave the saw in the timber and continue other work

Q113 LO6 · AC 6.3

Bore cutting is performed using:

A. The top of the bar in a controlled downward motion
B. The full length of the bar in a rapid thrust
C. The lower part of the bar nose, entering the wood to minimise kickback risk '
D. The chain at idle speed only

Q114 LO6 · AC 6.3

Why is bore cutting considered a higher-risk technique?

A. It consumes significantly more fuel
B. It requires a longer bar than standard cross-cutting
C. It is only a risk in wet or icy conditions
D. It uses the bar nose area where kickback is most likely to be initiated '

Q115 LO6 · AC 6.3

Before performing a bore cut, the operator must:

A. Assess wood tension and compression and confirm a clear escape route '
B. Loosen the chain for greater flexibility
C. Set the engine to idle speed for safety
D. Ensure the bar is shorter than 35cm

Q116 LO6 · AC 6.3

Bore cutting is most commonly used for:

A. Routine cross-cutting of felled timber
B. Sharpening the chain in the field
C. Cutting timber where access is limited from one side '
D. Felling trees with a diameter greater than bar length

Q117 LO6 · AC 6.4

When timber diameter exceeds the guide bar length, the correct cross-cutting technique is:

A. Fit a longer bar before attempting the cut
B. Cut at an angle to reach the centre of the log
C. Do not attempt — the timber cannot be safely cut
D. Cut from one side, then reposition and cut from the opposite side '

Q118 LO6 · AC 6.4

The key additional hazard when working with large diameter timber is:

A. Roll hazard — large logs can roll unexpectedly during or after cutting '
B. The increased weight of the chainsaw
C. Increased noise levels from the larger log
D. Excessive sawdust production

Q119 LO6 · AC 6.4

Tensioned timber creates a specific risk of:

A. The wood burning during the cutting process
B. The chain becoming magnetised
C. Sudden and unpredictable movement of log sections endangering the operator '
D. Fuel contamination from the timber

Q120 LO6 · AC 6.4

Before cutting oversized or tensioned timber, the operator must:

- **A. Thoroughly assess the timber, identify tension/compression zones, plan all cuts, and confirm escape route '**
- B. Begin cutting immediately to release the tension safely
- C. Ask a bystander to watch from a safe distance
- D. Wait 10 minutes after the timber has come to rest before cutting

Q121 LO3 · AC 3.2

What is the primary advantage of a battery-powered chainsaw over a petrol chainsaw?

- A. Greater cutting power for heavy timber
- B. Longer run time without stopping
- C. Cheaper to purchase initially
- **D. Zero exhaust emissions and lower noise levels, making it suitable for indoor or urban environments '**

Q122 LO3 · AC 3.2

How should a lithium-ion chainsaw battery be stored when not in use for an extended period?

- A. Fully discharged in a sealed plastic bag
- B. Fully charged in direct sunlight to maintain cell voltage
- **C. At approximately 40–60% charge in a cool, dry place away from extreme temperatures '**
- D. Submerged in oil to prevent corrosion

Q123 LO3 · AC 3.2

Which of the following is a key limitation of battery-powered chainsaws compared to petrol models?

- A. They cannot be fitted with a chain brake
- B. They require more PPE than petrol chainsaws
- **C. Run time is limited by battery capacity, requiring charged spare batteries for prolonged use '**
- D. They are heavier due to the engine block

Q124 LO3 · AC 3.2

What safety check is unique to a battery-powered chainsaw before use?

- **A. Verify the battery charge level and inspect the battery and terminals for damage before fitting '**
- B. Check the spark plug gap
- C. Check the fuel-to-oil mix ratio
- D. Test the recoil starter mechanism

Q125 LO1 · AC 1.1

What condition is caused by prolonged exposure to hand-arm vibration from chainsaws?

- A. Carpal tunnel syndrome only
- **B. Hand-Arm Vibration Syndrome (HAVS) '**
- C. Repetitive strain injury (RSI)
- D. Arthritis

Q126 LO1 · AC 1.1

Which of the following is an early symptom of Hand-Arm Vibration Syndrome (HAVS) that should be reported to an employer?

- A. Improved grip strength after extended use
- **B. Tingling, numbness or whitening of fingers, particularly in cold conditions '**
- C. Mild muscle soreness in the upper back only
- D. Increased sensitivity to heat in the palms

Q127 LO1 · AC 1.1

What is the MAIN risk to hearing from chainsaw operation?

- A. Temporary hearing loss only, which always recovers
- **B. Permanent noise-induced hearing loss '**
- C. Tinnitus that resolves after rest
- D. Pressure damage from exhaust fumes

Q128 LO1 · AC 1.1

Chainsaw exhaust fumes contain which dangerous gas that can build up in poorly ventilated areas?

A. Carbon dioxide (CO ,)
B. Nitrogen dioxide (NO ,)
C. Carbon monoxide (CO) '
D. Sulphur dioxide (SO ,)

Q129 LO1 · AC 1.2

What standard must a safety helmet worn during chainsaw work meet in the UK?

A. BS EN 388
B. BS EN 397 '
C. BS EN 471
D. BS EN 166

Q130 LO1 · AC 1.2

When must hearing protection be worn during chainsaw operation?

A. Only when working within 10 metres of other people
B. Only on tasks lasting over 30 minutes
C. At all times when the saw is running — chainsaw noise routinely exceeds safe exposure limits '
D. Only when instructed by a supervisor on site

Q131 LO2 · AC 2.2

Under RIDDOR 2013, which type of accident at work must be reported to the HSE?

A. Any bruise or scrape
B. Over-three-day injuries only
C. Specified injuries, dangerous occurrences, and deaths '
D. Minor cuts requiring no medical treatment

Q132 LO2 · AC 2.2

What type of fire extinguisher is safest to use on a chainsaw fuel fire?

A. Water
B. CO , or dry powder '
C. Foam only
D. Wet chemical

Q133 LO2 · AC 2.2

If a chainsaw operator collapses due to suspected carbon monoxide poisoning, what is the FIRST action?

A. Give the casualty water
B. Move the casualty to fresh air immediately '
C. Apply a tourniquet
D. Administer pure oxygen from a canister

Q134 LO2 · AC 2.2

What information should be written on a site emergency card before work begins?

A. Operator's personal insurance details
B. Grid reference or address, nearest hospital, and emergency contact numbers '
C. The operator's next of kin only
D. The local authority planning reference

Q135 LO3 · AC 3.3

What does the right-hand rear guard (back hand guard) on a chainsaw protect the operator from?

A. Fuel spray during refuelling
B. Injury to the right hand if the chain breaks or derails from the guide bar '
C. Exhaust heat on the throttle hand
D. Accidental contact with the front hand guard

Q136 LO3 · AC 3.3

Which chain brake activation mechanism fires automatically during kickback WITHOUT requiring physical contact with the front hand guard?

A. The throttle interlock mechanism
B. The rear hand guard pressure sensor
C. The inertia-activated brake, triggered by rapid rotation of the saw body '
D. The decompression valve releasing

Q137 LO3 · AC 3.3

How often should the chain brake function be checked as part of routine chainsaw operation?

A. Once per week before the first job of the week
B. Before every work session as part of pre-use checks '
C. Only after a suspected kickback incident has occurred
D. During the annual service only

Q138 LO3 · AC 3.3

If anti-vibration (AV) mounts are found to be cracked or hardened during a chainsaw inspection, the correct action is:

A. Continue using the saw but limit each session to 30 minutes
B. Apply silicone sealant to the cracks and continue monitoring
C. Do not use the saw — arrange replacement of AV mounts before further operation '
D. Only replace the mounts during the next scheduled annual service

Q139 LO3 · AC 3.2

What does the depth gauge (raker) on a chainsaw chain control?

A. The angle at which the file is held during sharpening
B. The speed at which the chain runs around the bar
C. The depth of cut — how much material each cutter tooth removes with each pass '
D. The tension of the chain on the guide bar

Q140 LO3 · AC 3.2

If depth gauges are set too low on a chainsaw chain, what is the most likely effect?

A. The chain will not cut
B. Increased vibration and kickback risk '
C. The chain will slip off the bar
D. Reduced fuel consumption

Q141 LO4 · AC 4.1

How should chain tension be checked on a chainsaw?

A. The chain should sag visibly below the bar
B. The chain should sit flush with the bar groove and snap back when pulled '
C. The chain should be tight enough that it cannot be moved by hand
D. Chain tension should only be checked when the chain is cold

Q142 LO4 · AC 4.1

What does the drive link on a chainsaw chain do?

A. Connects the cutting teeth to the tie straps
B. Fits into the bar groove and is driven by the sprocket '
C. Provides the depth gauge function
D. Attaches the chain to the clutch drum

Q143 LO4 · AC 4.1

Why must the chainsaw bar groove be cleaned regularly?

A. To improve the appearance of the bar
B. To prevent sawdust build-up that can cause the chain to jam or the bar to overheat '
C. To allow the chain to run faster
D. To prevent rust forming on the bar tip

Q144 LO4 · AC 4.1

In which direction should a correctly fitted chainsaw chain travel along the top of the bar?

- **A. Away from the operator (towards the bar tip) '**
- B. Towards the operator
- C. Either direction is acceptable
- D. The direction depends on wood species

Q145 LO5 · AC 5.1

As part of a pre-start inspection, how should the operator confirm the bar oil tank has been filled?

- A. Remove the bar and tip the saw to check for oil flow
- B. Check the oil cap is tight — no further check required
- **C. Check the oil tank level visually or with a dipstick and fill before starting '**
- D. Run the saw for 30 seconds first to prime the oiling system

Q146 LO5 · AC 5.1

During a pre-start guide bar inspection, which condition would require the bar to be taken out of service?

- A. Minor surface rust on the bar rails
- B. Sawdust packed in the oil feed hole (cleaned before use)
- **C. A cracked or bent bar, or rails worn beyond manufacturer specification '**
- D. A bar sprocket nose making occasional clicking noises

Q147 LO5 · AC 5.2

When performing a warm start on a chainsaw that has recently been running, the choke should be:

- A. Set to half-choke to enrich the warm fuel mixture
- B. Fully closed until the engine fires on the first pull
- **C. Left in the open (run) position — no choke is needed for a warm engine '**
- D. Adjusted to the manufacturer's warm-start position on the carburettor

Q148 LO5 · AC 5.2

During a pre-use check, why must the operator confirm that all controls return to their neutral position when released?

- A. To confirm fuel is reaching the carburettor correctly
- **B. To prevent the chain running unintentionally, reducing injury risk during operation '**
- C. To check the chain brake is correctly disengaged
- D. To verify engine temperature is within normal range

Q149 LO6 · AC 6.2

A log is resting on two supports with the cut point in the middle. Where is the compression zone, and how should the cut be started?

- A. Compression at the bottom — start cutting from below and cut upward
- **B. Compression at the top — start from below (tension side) so the kerf opens rather than pinches '**
- C. No compression zone exists when both ends are supported
- D. Compression at the top — start from the top to cut through it first

Q150 LO6 · AC 6.2

A log is cantilevered — supported at one end only, with the cut near the free overhanging end. Where is the compression zone?

- A. At the top of the log — start from above and cut downward through the tension zone last
- **B. At the bottom of the log — start from above (tension side) to avoid bar pinch from below '**
- C. There is no compression in an overhanging log
- D. At the top of the log — start from below so compression holds the kerf open

Q151 LO1 · AC 1.1, AC 1.2

Under the Manual Handling Operations Regulations 1992 (MHOR), where manual handling involving a risk of injury cannot be avoided, what must an employer do?

- A. Issue all workers with a manual handling certificate
- **B. Make a suitable assessment of the manual handling task and take appropriate steps to reduce the risk of injury '**
- C. Prohibit manual handling activities entirely on site
- D. Ensure a written permit to lift is issued before any log is moved

SECTION 7

Supplementary Oral & Practical Mock Questions

79 Questions · Formative Practice Only — Not Formally Assessed

Important — these questions do not contribute to the pass/fail outcome

These 79 questions are oral and practical preparation aids. They do NOT form part of the summative examination, do NOT appear on the certificate, and are NOT formally assessed. Within the app, learners practise them via an AI-assisted voice or text feature that provides formative feedback.

M1 Oral / Practical · Formative Only

What are the 5 steps to risk assessment?

Key points:

- [object Object]

M2 Oral / Practical · Formative Only

Give me at least three hazards with the site and their control measures.

Key points:

- [object Object]

M3 Oral / Practical · Formative Only

Give me at least three hazards with the chainsaw and their control measures.

Key points:

- [object Object]

M4 Oral / Practical · Formative Only

Give me at least three hazards with the job to be done and the control measures.

Key points:

- [object Object]

M5 Oral / Practical · Formative Only

What emergency information do I need to know for the site?

Key points:

- [object Object]

M6 Oral / Practical · Formative Only

What is the Health and Safety at Work Act all about?

Key points:

- [object Object]

M7 Oral / Practical · Formative Only

Under PUWER, what does it say about the equipment?

Key points:

- [object Object]

M8 Oral / Practical · Formative Only

Who provides industry guidance for tree work?

Key points:

- [object Object]

M9 Oral / Practical · Formative Only

Why is it important to maintain the saw to the manufacturer's specifications?

Key points:

- [object Object]

M10 Oral / Practical · Formative Only

The manufacturer has installed safety features on the saw — run through where they are and what they do.

Key points:

- [object Object]

M11 Oral / Practical · Formative Only

If you find one of the safety features is broken or missing, what will you do?

Key points:

- [object Object]

M12 Oral / Practical · Formative Only

What are the advantages of using battery chainsaws?

Key points:

- [object Object]

M13 Oral / Practical · Formative Only

What are the disadvantages of using battery chainsaws?

Key points:

- [object Object]

M14 Oral / Practical · Formative Only

What maintenance might you do with the batteries, the saw and charger?

Key points:

- [object Object]

M15 Oral / Practical · Formative Only

Can you take your top cover off your chainsaw please.

Key points:

- [object Object]

M16 Oral / Practical · Formative Only

So what's this?

Key points:

- [object Object]

M17 Oral / Practical · Formative Only

What does it do? (air filter)

Key points:

- [object Object]

M18 Oral / Practical · Formative Only

How do you maintain it? (air filter)

Key points:

- [object Object]

M19 Oral / Practical · Formative Only

So what's this? (spark plug)

Key points:

- [object Object]

M20 Oral / Practical · Formative Only

What does it do? (spark plug)

Key points:

- [object Object]

M21 Oral / Practical · Formative Only

What are you looking for and how would you maintain it? (spark plug)

Key points:
- [object Object]

M22 Oral / Practical · Formative Only

In conjunction with the flywheel, what do these cylinder fins help do?

Key points:
- [object Object]

M23 Oral / Practical · Formative Only

How do you maintain the cooling system?

Key points:
- [object Object]

M24 Oral / Practical · Formative Only

So what's this? (exhaust)

Key points:
- [object Object]

M25 Oral / Practical · Formative Only

What does it do? (exhaust)

Key points:
- [object Object]

M26 Oral / Practical · Formative Only

How do you maintain it? (exhaust)

Key points:
- [object Object]

M27 Oral / Practical · Formative Only

What is in here that you need to maintain and how would you do it? (fuel tank)

Key points:
- [object Object]

M28 Oral / Practical · Formative Only

What is in here that you need to maintain and how would you do it? (oil tank)

Key points:
- [object Object]

M29 Oral / Practical · Formative Only

This is the recoil housing — can you take it off and de-tension the pull cord please.

Key points:
- [object Object]

M30 Oral / Practical · Formative Only

Where would you expect to find damage? (recoil starter)

Key points:
- [object Object]

M31 Oral / Practical · Formative Only

Now can you take off the chain and bar side cover please.

Key points:
- [object Object]

M32

Oral / Practical · Formative Only

So what's this? (sprocket)

Key points:

- [object Object]

M33

Oral / Practical · Formative Only

What does it do? (sprocket)

Key points:

- [object Object]

M34

Oral / Practical · Formative Only

How do you check for any wear or damage? (sprocket)

Key points:

- [object Object]

M35

Oral / Practical · Formative Only

If you need to replace the sprocket, how are you going to do that?

Key points:

- [object Object]

M36

Oral / Practical · Formative Only

Now on to the chain brake — where would you find it on your saw?

Key points:

- [object Object]

M37

Oral / Practical · Formative Only

What does the chain brake do?

Key points:

- [object Object]

M38

Oral / Practical · Formative Only

How do you check the chain brake for any wear or damage?

Key points:

- [object Object]

M39

Oral / Practical · Formative Only

This is your guidebar — what signs of wear or damage are you looking for?

Key points:

- [object Object]

M40

Oral / Practical · Formative Only

If the bar shows excessive wear or damage, what problems might you get?

Key points:

- [object Object]

M41

Oral / Practical · Formative Only

How are you going to maintain your bar?

Key points:

- [object Object]

M42

Oral / Practical · Formative Only

Can you reassemble your saw please.

Key points:

- [object Object]

M43 Oral / Practical · Formative Only

When you're ready, can you clamp your saw into a vice please.

Key points:

- [object Object]

M44 Oral / Practical · Formative Only

How do you identify your chain?

Key points:

- [object Object]

M45 Oral / Practical · Formative Only

What information do you need to replace your chain?

Key points:

- [object Object]

M46 Oral / Practical · Formative Only

What cutter profile do you have on your chain?

Key points:

- [object Object]

M47 Oral / Practical · Formative Only

What is the other main type of cutter profile?

Key points:

- [object Object]

M48 Oral / Practical · Formative Only

And what are their different uses?

Key points:

- [object Object]

M49 Oral / Practical · Formative Only

What top plate angle are you filing at?

Key points:

- [object Object]

M50 Oral / Practical · Formative Only

And what file size are you using?

Key points:

- [object Object]

M51 Oral / Practical · Formative Only

So where are you going to start sharpening?

Key points:

- [object Object]

M52 Oral / Practical · Formative Only

So what's this? (depth gauge / raker)

Key points:

- [object Object]

M53 Oral / Practical · Formative Only

What does it do? (depth gauge)

Key points:

- [object Object]

M54 Oral / Practical · Formative Only

Why does the depth gauge need to be set correctly, and what's the danger in setting it too low?

Key points:

- [object Object]

M55 Oral / Practical · Formative Only

Why do you want to keep all your cutters the same size and the angles all the same?

Key points:

- [object Object]

M56 Oral / Practical · Formative Only

What problems might you get if you use a really worn or badly sharpened chain?

Key points:

- [object Object]

M57 Oral / Practical · Formative Only

How do you dispose of your contaminated chainsaw waste and your litter?

Key points:

- [object Object]

M58 Oral / Practical · Formative Only

What is the safe working distance between operators when cross cutting?

Key points:

- [object Object]

M59 Oral / Practical · Formative Only

What bio-security measures might you put in place?

Key points:

- [object Object]

M60 Oral / Practical · Formative Only

What environmental factors do you need to consider?

Key points:

- [object Object]

M61 Oral / Practical · Formative Only

In this picture, where is the compression found?

Key points:

- [object Object]

M62 Oral / Practical · Formative Only

And in this picture, where is the compression?

Key points:

- [object Object]

M63 Oral / Practical · Formative Only

What are you going to do if you get your saw stuck or pinched?

Key points:

- [object Object]

M64 Oral / Practical · Formative Only

How are you going to cross cut a piece of timber that's slightly larger than your guidebar?

Key points:

- [object Object]

M65 Oral / Practical · Formative Only

How are you going to cut timber that's under high tension?

Key points:

- [object Object]

M66 Oral / Practical · Formative Only

When would you use a bore cut when cross cutting timber?

Key points:

- [object Object]

M67 Oral / Practical · Formative Only

Once you've cut your timber, how are you going to move it safely?

Key points:

- [object Object]

M68 Oral / Practical · Formative Only

How do you make moving and working more ergonomic?

Key points:

- [object Object]

M69 Oral / Practical · Formative Only

How are you going to make sure your timber stacks are stable?

Key points:

- [object Object]

M70 Oral / Practical · Formative Only

What else do you need to think about when stacking lots of different timber?

Key points:

- [object Object]

M71 Oral / Practical · Formative Only

Before you start your saw, what checks do you need to do?

Key points:

- [object Object]

M72 Oral / Practical · Formative Only

What functional checks do you need to perform before you actually use your saw?

Key points:

- [object Object]

M73 Oral / Practical · Formative Only

With all checks done and everything in good condition and working correctly, you are ready to undertake the practical assessment.

Key points:

- [object Object]

M74 Oral / Practical · Formative Only

For the cutting part of the assessment you will need to demonstrate multiple cuts under tension and compression and perform bore cuts.

Key points:

- [object Object]

M75 Oral / Practical · Formative Only

Good luck.

Key points:

- [object Object]

M76

Oral / Practical · Formative Only

What bit of the bar do you need to avoid cutting with?

Key points:

- [object Object]

M77

Oral / Practical · Formative Only

What PPE do you need when using and operating a chainsaw?

Key points:

- [object Object]

M78

Oral / Practical · Formative Only

What do the Manual Handling Operations Regulations 1992 (MHOR) require you to do before manually moving heavy timber on site?

Key points:

- [object Object]

M79

Oral / Practical · Formative Only

How do the Manual Handling Operations Regulations 1992 apply to log handling and timber stacking?

Key points:

- [object Object]

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